Sahrdaya College of Engineering and Technology, Kodakara

Answer Key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SIXTH SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2022

Course Code: EET304

Course Name: POWER SYSTEMS II

PART A

Answer all questions. Each question carries 3 marks

Q1 A 50MVA, 11 kV, three phase synchronous generator was subjected to different types of faults. The fault currents are as follows: LG fault: 4130 A, LL fault 2590 A, 3 Phase fault: 1870 A. The generator neutral is solidly grounded. Find the p.u values of sequence reactances of the generator.

Ans LG Fault

$$I_f = I_a = 3\,I_{a1}$$

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 + Z_0}$$

LL Fault

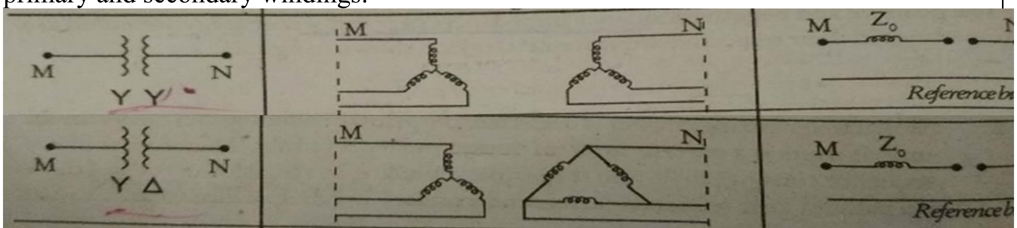
$$I_{a1} = \frac{E_a}{Z_1 + Z_2}$$

$$I_f = I_b = I_{a1}\,(a^2 - a)$$

- X1=3.396ohms-1 mark
- X2=0.851ohms-1 mark
- Xo=0.366 ohms-1 mark

Q2 Draw the zero sequence networks for a) Y-Y b) Y-Δ connected transformers.

Ans The zero sequence impedance of the 3ϕ transformer depends on the connection of the primary and secondary windings.



Q3 Explain the terms with respect to Gauss Siedel method a) Acceleration factor b) Convergence criteria c) Handling of PV buses.

Ans

In the Gauss-Seidel method, a large number of the iteration is required to arrive at the specified convergence. The rate of convergence can be increased by the use of the acceleration factor to the solution obtained after each iteration. The Acceleration factor is a multiplier that enhances correction between the values of voltage in two successive iterations. Convergence in the Gauss Seidel Method can sometimes be speeded up by the use of the acceleration factor. For the ith bus, the accelerated value of voltage at the (r + 1)th iteration is given by

$$V_i^{(r+1)}\ (\text{accelerated}) = V_i^{(r)} + \alpha(V_i^{(r+1)} - V_i^{(r)})$$

where α is a real number called the acceleration factor. A suitable value of α for any system can be obtained by trial load flow studies. A generally recommended value is α = 1.6. A wrong choice of cc may indeed slow down convergence or even cause the method to diverge. This concludes the load flow analysis for the case of PQ buses only.

Handling of PV buses

Temporarily set $|V_p^k| = |V_p|_{spec}$ and phase of V_p^k as the k^{th} iteration value if the bus-p is a generator bus where $|V_p|_{spec}$ is the specified magnitude of voltage for bus-p. Then calculate the reactive power of the generator bus using the following equation.

$$Q^{k+1}_{p,cal} = (-1)\times \mathrm{Im}\left\{(V_p^k)^* \left[\sum_{q=1}^{p-1} Y_{pq}V_q^{k+1} + \sum_{q=p}^{n} Y_{pq}V_q^k\right]\right\}$$

(Here "Im" stands for "Imaginary part of")

The calculated reactive power may be within specified limits or it may violate the limits.

If the calculated reactive power is within the specified limits then consider this bus as generator bus and set $Q_p = Q^{k+1}_{p,cal}$ for this iteration and go to step-8.

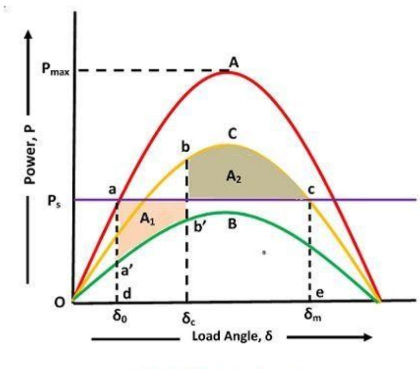
If the calculated reactive power violates the specified limit for reactive power then treat this bus as a load bus. The magnitude of the reactive power at this bus will correspond to the limit it has violated.

i.e., if $Q^{k+1}_{p,cal} < Q_{p,min}$ then $Q_p = Q_{p,min}$

(or) $Q^{k+1}_{p,cal} > Q_{p,max}$ then $Q_p = Q_{p,max}$

Since the bus is treated as load bus, take actual value of V_p^k for $(k+1)^{th}$, iteration. i.e., $|V_p^k|$ need not be replaced by $|V_p|_{spec}$ when the generator bus is treated as load bus. Go to step-9.

		For generator bus the magnitude of voltage does not change and so for all iterations the magnitude of bus voltage is the specified value. The phase of the bus voltage can be calculated as shown below. $$V_{p,temp}^{k+1} = \frac{1}{Y_{pp}}\left[\frac{P_p - jQ_p}{(V_p^k)^*} - \sum_{q=1}^{p-1} Y_{pq} V_q^{k+1} - \sum_{q=p+1}^{n} Y_{pq} V_q^k\right]$$ $$\delta_p^{k+1} = \tan^{-1}\left[\frac{\text{Im. part of } V_{p,\,temp}^{k+1}}{\text{Real part of } V_{p,\,temp}^{k+1}}\right]$$ *Now, the (k+1)th iteration voltage of the generator bus is given by* $$V_p^{k+1} = \lvert V_p \rvert_{spec} \angle \delta_p^{k+1}$$ *where* $\lvert V_p \rvert_{spec}$ *= magnitude of specified bus voltage.* *After calculating* V_p^{k+1} *for generator bus go to step-11.*
	Q4	State the static load flow problem.
	Ans	The load flow problem consists of finding the set of voltages: magnitude and angle, which, together with the network impedances, produces the load flows that are known to be correct at the system terminals. To start, we view the power system as being a collection of buses, connected together by lines. $$Y_{Bus}V = I$$ $$I_p = Y_{p1}V_1 + Y_{p2}V_2 + \ldots + Y_{pp}V_p + \ldots Y_{pn}V_n$$ $$\therefore I_p = \sum_{q=1}^{p-1} Y_{pq}V_q + Y_{pp}V_p + \sum_{q=p+1}^{n} Y_{pq}V_q$$ Let S_p = Complex power of bus-p P_p = Real power of bus-p Q_p = Reactive power of bus-p Now, $S_p = P_p + jQ_p$ Also, $S_p = V_p I_p^*$ $$\therefore V_p I_p^* = P_p + jQ_p$$ $$(V_p I_p^*)^* = (P_p + jQ_p)^*$$ $$V_p^* I_p = P_p - jQ_p$$ $$\therefore I_p = \frac{P_p - jQ_p}{V_p^*}$$

		$$\sum_{q=1}^{p-1} Y_{pq}V_q + Y_{pp}V_p + \sum_{q=p+1}^{n} Y_{pq}V_q = \frac{P_p - jQ_p}{V_p^*}$$
	Q5	Differentiate between steady state and transient stability?
		• Steady-state stability refers to the ability of the various machines to regain and maintain synchronism after a small and slow disturbance, such as a gradual change in load. Transient stability is stability after a sudden large disturbance such as a fault, loss of a generator, a switching operation, and a sudden load change. • Steady State Stability is defined as the stability of system under slow or gradual change in load. It is assumed that the rate of change of load is comparatively low when compared with the rate of change of field flux of machine/generator in response to change in load. • Transient State Stability is capability of system to be stable when there is a sudden or large change in system condition like load, fault at a point, change is mechanical input to the rotor shaft of Generator etc.
	Q6	What is critical clearance time and critical clearance angle? What is their significance in stability studies?
	Ans	The critical clearing angle is defined as the maximum change in the load angle curve before clearing the fault without loss of synchronism. In other words, when the fault occurs in the system the load angle curve begin to increase, and the system becomes unstable. The angle at which the fault becomes clear and the system becomes stable is called critical clearing angle. When the initial load is given, then there is a critical clearing angle, and if the actual clearing angle exceeds a critical clearing angle, the system becomes unstable otherwise it is stable. Let the curve A represents the power angle curve for a healthy condition; curve B represents the power angle curve for faulty condition and curve C represents the power angle curve after isolation of fault as shown below.  Critical Clearing Angle For transient stability limit, two areas A1 = A2 or in other words the area under curve adec (rectangle) is equal to the area under the curve da'b'bce.
	Q7	What is SCADA? Give its purpose.
	Ans	Supervisory control and data acquisition (SCADA) is a system of software and hardware elements that allows industrial organizations to: • Control industrial processes locally or at remote locations • Monitor, gather, and process real-time data

		• Directly interact with devices such as sensors, valves, pumps, motors, and more through human-machine interface (HMI) software • Record events into a log file SCADA systems are crucial for industrial organizations since they help to maintain efficiency, process data for smarter decisions, and communicate system issues to help mitigate downtime. The basic SCADA architecture begins with programmable logic controllers (PLCs) or remote terminal units (RTUs). PLCs and RTUs are microcomputers that communicate with an array of objects such as factory machines, HMIs, sensors, and end devices, and then route the information from those objects to computers with SCADA software. The SCADA software processes, distributes, and displays the data, helping operators and other employees analyze the data and make important decisions.
	Q8	What is control area and ACE?
	Ans	Control Area means an electric system or combination of electric power systems to which a common automatic generation control scheme is applied in order to: (1) match, at all times, the power output of the Generators within the electric power system (s) and capacity and energy purchased from entities outside the electric power system (s), with the Load within the electric power system (s); (2) maintain scheduled interchange with other Control Areas, within the limits of Good Utility Practice. A suitable linear combination of frequency and tie line power changes for area i, is known as the area control error. Actually ACE is the difference between scheduled and actual electrical generation within a control area on the power grid, taking frequency bias into account. Area control error, commonly called ACE, is a quantity used in operating bulk electric systems. ACE is defined as the instantaneous difference between a balancing authority 's net actual and scheduled interchange with all adjacent interconnected balancing authority areas. in a multi-area power system, in addition to regulating area frequency, the supplementary control should maintain the net interchange power with neighboring areas at scheduled values. This is generally accomplished by adding a tie line flow deviation to the frequency deviation in the supplementary feedback loop. A suitable linear combination of frequency and tie line power changes for area i , is known as the area control error. Actually ACE is the difference between scheduled and actual electrical generation within a control area on the power grid, taking frequency bias into account.
	Q9	Examine the following: 1) Reasons for maintaining spinning reserve 2) Significance of start-up cost
	Ans	1) Spinning reserve must be maintained so that failure of one or more units does not cause too far a drop in system frequency, i.e., if one unit fails, there must be ample reserve on the other units to make up for the loss in a specified time period. 2) Because of temperature and pressure of a thermal unit that must be moved slowly, a certain amount of energy must be moved slowly, a certain amount of energy must be expended to bring the unit on-line, and it is brought into the UC problem as a start-up cost. The start-up cost may vary from a maximum cold-start value to a very small value if the unit was only turned off recently and is still relatively close to the operating temperature.
	Q10	Derive the condition for economic dispatch when losses are considered.
	Ans	The optimal load dispatch problem including transmission losses is defined as $$\text{Min } F_T = \sum_{n=1}^{n} F_n \qquad (1)$$

$$\text{Subject to } P_D + P_L - \sum_{n=1}^{n} P_n = 0 \qquad (2)$$

P_L – Total system loss which is assumed to be a function of generation
Making use of Lagrangian multiplier λ, the auxiliary function is given by

$$F = F_T + \lambda(P_D + P_L - \sum_{n=1}^{n} P_n)$$

The partial differential of this expression when equated to zero gives the condition for optimal load dispatch ie;

$$\frac{\partial F}{\partial P_n} = \frac{\partial F_T}{\partial P_n} + \lambda\left(\frac{\partial P_L}{\partial P_n} - 1\right) = 0$$

Since $F_T = F_1 + F_2 + \ldots\ldots\ldots + F_n$

$$\frac{\partial F_T}{\partial P_n} = \frac{dF_n}{dP_n}$$

$$\frac{dF_n}{dP_n} + \lambda\frac{\partial P_L}{\partial P_n} = \lambda \qquad (3)$$

Here, $\frac{\partial P_L}{\partial P_n}$ - the transmission loss at plant n

λ - Incremental cost of received power in Rs per MWhr

- The equation (3) is a set of n equations with (n+1) unknowns. Here n generations are unknown & λ is also unknown.
- These equations are known as coordination equations because they coordinate the incremental transmission losses with the incremental cost of production
- To solve these equations the loss formula equation (4) is expressed in terms of generations & is expressed as

$$P_L = \sum_m \sum_n P_m B_{mn} P_n \qquad (4)$$

P_m & P_n- source loadings
B_{mn} -the transmission loss coefficients
The formula is derived under the following assumptions:

1. The equivalent load current at any bus remains a constant complex fraction of the total equivalent load current
2. The generator bus voltage magnitudes & angles are constant
3. The power factor of each source is constant

The solution of coordination equation (3) requires the calculation of $\frac{\partial P_L}{\partial P_n}$ which is obtained from equation (4) as

$$\frac{\partial P_L}{\partial P_n} = 2\sum_m B_{mn} P_m \qquad (5)$$

$$\frac{dF_n}{dP_n} = F_{nn} P_n + f_n$$

The coordination equation can be rewritten as

$F_{nn} P_n + f_n + \lambda \sum_m 2B_{mn} P_m = \lambda \quad (6)$

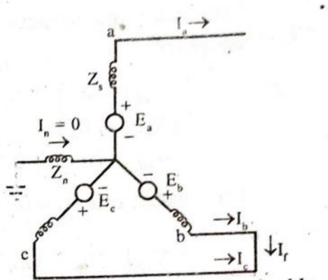
Collecting all coefficients of P_n,
$P_n(F_{nn} + 2\lambda B_{nn}) = -\lambda\left(\sum_{m \neq n} 2B_{mn} P_m\right) - f_n + \lambda$

		Solving for P_n $P_n = \dfrac{1 - \frac{f_n}{\lambda} - \sum_{m \neq n} 2B_{mn}P_m}{\frac{F_{nn}}{\lambda} + 2B_{nn}} \qquad (7)$

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

	Q11	a) Derive the expression for fault current in a line to line fault on unloaded generator. Draw an equivalent network showing the inter connection of networks to simulate a line to line fault. b) What are current limiting reactors? Give its function and location.
	ans	The circuit diagram for a line-to-line fault on an unloaded , Y-connected generator is shown in fig 4.6. Here it is assumed that phase-b and phase-c are shorted. Now the fault current $I_f = I_b = -I_c$. Since the generator is unloaded the current in phase-a is zero. The conditions at the fault are expressed by the following equations $V_b = V_c \;;\; I_a = 0 \;;$ $I_b + I_c = 0 \Rightarrow I_b = -I_c \quad .....(4.29)$ With $V_c = V_b$, the symmetrical components of voltage are given by $\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} V_a \\ V_b \\ V_b \end{bmatrix} \quad .....(4.30)$  *Fi 4.6 : Circuit diagram for double li to ground fault between phase b c in an unloaded generator*

On multiplying row 2 & 3 of equ(4.30) we get,

$$V_{a1} = \frac{1}{3}(V_a + aV_b + a^2V_b)$$

$$V_{a2} = \frac{1}{3}(V_a + a^2V_b + aV_b)$$

From equations (4.31) and (4.32) we can write

$$V_{a1} = V_{a2}$$

The symmetrical components of current are given by

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

On substituting $I_b = -I_c$ and $I_a = 0$ in equ(4.34) we get

$$\begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} 0 \\ -I_c \\ I_c \end{bmatrix}$$

On multiplying equ(4.35) we get,

$$I_{a0} = \frac{1}{3}\left[-I_c + I_c\right] = 0$$

$$I_{a1} = \frac{1}{3}\left[-a\, I_c + a^2\, I_c\right]$$

$$I_{a2} = \frac{1}{3}\left[-a^2\, I_c + a\, I_c\right] = -\frac{1}{3}\left[-a\, I_c + a^2\, I_c\right]$$

From equations (4.36) and (4.37) we can write

$$I_{a2} = -I_{a1}$$

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} I_{a0} \\ I_{a1} \\ I_{a2} \end{bmatrix}$$

On substituting $I_{a0} = 0$ and $I_{a2} = -I_{a1}$ in equ (4.39) we get,

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \begin{bmatrix} 0 \\ E_a \\ 0 \end{bmatrix} - \begin{bmatrix} Z_0 & 0 & 0 \\ 0 & Z_1 & 0 \\ 0 & 0 & Z_2 \end{bmatrix} \begin{bmatrix} 0 \\ I_{a1} \\ -I_{a1} \end{bmatrix}$$

On multiplying equ (4.40) we get,

$$V_{a0} = 0$$

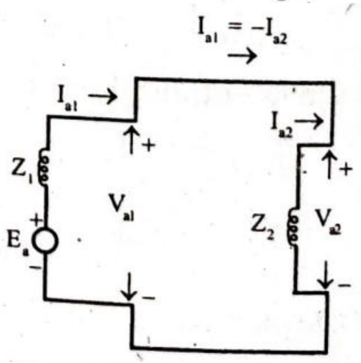
$$V_{a1} = E_a - Z_1 I_{a1}$$

$$V_{a2} = Z_2 I_{a1}$$

$$E_a - Z_1 I_{a1} = Z_2 I_{a1}$$

(or) $I_{a1} (Z_1 + Z_2) = E_a$

$$\therefore I_{a1} = \frac{E_a}{Z_1 + Z_2}$$



$$I_b = a^2 I_{a1} - a I_{a2} = I_{a1} (a^2 - a)$$

$$\therefore \text{Fault current, } I_f = I_b = I_{a1} (a^2 - a)$$

b) Current limiting reactors

A series reactor is an inductive coil connected in a circuit or system to limit the short circuit current. The short circuit current in a power system depends on the system generating capacity, voltage at the point of fault & the impedance between the generators & the fault point. A series reactor may be of fixed or switched type. A fixed reactor is permanently connected in the circuit. A switched reactor is one whose reactance can be varied so that at any time a desired amount of reactance can be connected in the system. The specifications for a reactor include the type, location (indoor or outdoor), reactance in ohm, KVA rating, rated voltage, rated current & frequency

a) Generator reactor:
   - It is connected in series with alternator
   - Modern alternators have high inherent reactance & therefore do not need any external reactor for limiting the current
   - They are still used for some old machines

b) Feeder reactor:
   - A reactor is connected in series with each feeder
   - Since the number of feeders is generally large, this connection entails the use of a large number of reactors
   - Since the reactors carry full load current, a constant power loss occurs

c) Bus bar reactor:
   - The reactors are inserted in between bus sections
   - During normal operation only a small power flows through the reactor & hence very small power loss takes place. During short circuit only that generator feeds the fault directly
   - The contribution of fault current by other generators is limited by the reactors

d) Tie bus-bar reactor:
   - The reactors are connected between the buses & a tie bar
   - This scheme has the advantage that the fault current is independent of the number of generators connected in the system

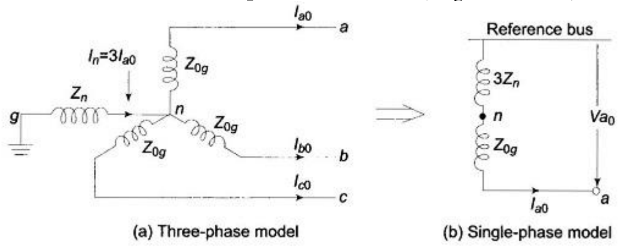
Q12

a) Give reasons a) The neutral grounding impedance Zn appears as 3Zn in the zero sequence equivalent circuit. b) Zero sequence currents cannot flow in the line currents of a delta connected system.

b) A synchronous generator and a synchronous motor are rated 25 MVA, 11kV and having sub transient reactance of 15% are connected through transformers and a transmission line. The transformers are rated 25MVA, 11/66 kV and 66/11kV respectively with leakage reactance of 10%. The line connecting them has a reactance of 10% on the base of 25MVA, 66kV. The motor is drawing 15 MW at 0.8pf lead and a terminal voltage of 10.6kV when a symmetrical 3phase fault occurs at the motor terminals. Find the sub transient current in the motor, generator and fault by using the internal voltages of the machines.

Ans

a) The flow of zero sequence currents creates three mmfs which are in time phase but are distributed in space phase by 120°. The resultant air gap field caused by zero sequence currents is therefore zero. Hence, the rotor windings present leakage reactance only to the flow of zero sequence currents ($Z_{0g} < Z_2 < Z_1$).



10.14 Zero sequence network of a synchronous machine

Zero sequence network models on a three- and single-phase basis are shown in Figs.

10.14a and b. In Fig. 10.14a, the current flowing in the impedance Z_n between neutral and ground is $I_n = 3I_{a0}$. The zero sequence voltage of terminal a with respect to ground, the reference bus, is therefore

$$V_{a0} = -3Z_nI_{a0} - Z_{0g}I_{a0} = -(3Z_n + Z_{0g})I_{a0} \quad (10.52)$$

where Z_{0g} is the zero sequence impedance per phase of the machine. Since the single-phase zero sequence network of Fig. 10.14b carries only per phase zero sequence current, its total zero sequence impedance must be

$$Z_0 = 3Z_n + Z_{0g} \quad (10.53)$$

In a balanced Y-connected system (with neutral ungrounded) the line current cannot have any zero sequence components since the neutral current is zero. In a balanced delta-connected system the line current cannot have any zero sequence components, however zero sequence currents can circulate in the closed delta winding. It is circulated in the phase windings of the delta connection as shown in the figure below.The zero sequence currents are produced due to the existence of zero sequence voltage. The above equation shows that there is no zero sequence current present in the delta connection, because of the absence of the return paths of these current.

b)

- Choose appropriate base values & determine the pre-fault condition reactance diagram of the given power system
- Calculate the pre-fault voltage at the fault point using the pre-fault current (load current). The pre-fault voltage at the fault point is the thevenin's voltage
- Determine the thevenin's impedance of the system at the fault point. To determine the thevenin's impedance, replace all the sources by zero value source & then reduce the resultant network to single equivalent impedance
- Draw the thevenin's equivalent at the fault point, F as shown in figure. Here the fault can be represented by a short circuit or by fault impedance

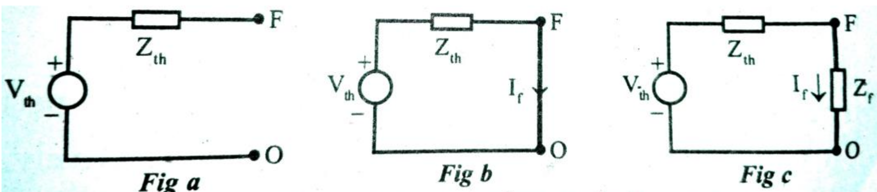


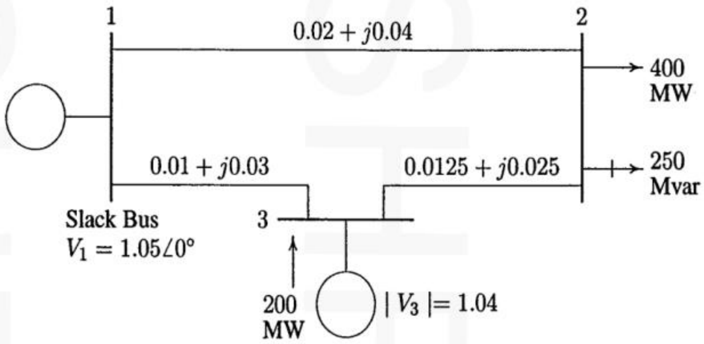
Fig 3.8 : Thevenin's equivalent at fault point

Now the p.u. value fault current I_f is given by

$$\text{p.u. value of fault current, } I_f = \frac{V_{th}}{Z_{th}}$$

- The fault current (ie; post fault current) in any part of the system is given by sum of pre-fault current & change in current due to fault
- The change in current due to fault can be estimated by connecting the thevenin's source with reversed polarity at the fault. Replace all other sources by zero value sources. Now the currents in various parts of the system are the change in currents due to fault.

$$I_g'' = I_1 + I_L$$

		$I_m'' = I_2 - I_L$ Prefault voltage=0.9636 pu and curret=0.7783<36.8 Ig”=816.4-j2197 Im’=-816.2-j9041.8 Fault current=-j11239A
	Module 2	
Q13	a)	Consider the one line diagram of a power system as shown in Fig.1.Line impedances are marked in per unit on a 100MVA base. Obtain the load flow solution using Gauss Siedel technique.  Fig.1 b) Write a brief note on fast decoupled load flow.
ans		The admittance matrix is given as $\mathrm{Y_{bus}} = \begin{vmatrix} y_{12}+y_{13} & -y_{12} & -y_{13} \\ -y_{21} & y_{21}+y_{23} & -y_{23} \\ -y_{31} & -y_{32} & y_{32}+y_{31} \end{vmatrix}$ Assume initial voltages to all buses $V_1\,(0) = 1.05\angle 0^0 = 1.05+j0$ p.u For bus 2 $V_p^{k+1} = \frac{1}{Y_{pp}}\left[\frac{P_p - jQ_p}{(V_p^k)^*} - \sum_{q=1}^{p-1} Y_{pq}V_q^{k+1} - \sum_{q=p+1}^{n} Y_{pq}V_q^k\right]$ For bus 3 $Q_{p,cal}^{k+1} = (-1)\times \mathrm{Im}\left\{(V_p^k)^*\left[\sum_{q=1}^{p-1} Y_{pq}V_q^{k+1} + \sum_{q=p}^{n} Y_{pq}V_q^k\right]\right\}$

$$V_{p,temp}^{k+1} = \frac{1}{Y_{pp}}\left[\frac{P_p - jQ_p}{(V_p^k)^*} - \sum_{q=1}^{p-1} Y_{pq} V_q^{k+1} - \sum_{q=p+1}^{n} Y_{pq} V_q^k\right]$$

$$\delta_p^{k+1} = \tan^{-1}\left[\frac{\text{Im. part of } V_{p,\,temp}^{k+1}}{\text{Real part of } V_{p,\,temp}^{k+1}}\right]$$

Now, the (k+1)th iteration voltage of the generator bus is given by

$$V_p^{k+1} = |V_p|_{spec} \angle \delta_p^{k+1}$$

where $|V_p|_{spec}$ *= magnitude of specified bus voltage.*

V_2^1= 0.9746-j0.0423

Q_3^1=1.16

V_3^1= 1.0399-j0.005

b) The fast decoupled power flow method is a very fast and efficient method of obtaining power flow problem solution. In this method, both, the speeds as well as the sparsity are exploited. This method exploits the property of the power system where in MW flow-voltage angle and MVAR flow-voltage magnitude are loosely coupled. In other words a small change in the magnitude of the bus voltage does not affect the real power flow at the bus and similarly a small change in phase angle of the bus voltage has hardly any effect on reactive power flow. Because of this loose physical interaction between MW and MVAR flows in a power system, the MW- δ and MVAR-V calculations can be decoupled. This decoupling results in a very simple, fast and reliable algorithm. As we know, the sparsity feature of admittance matrix minimizes the computer memory requirements and results in faster computations.

$$\begin{bmatrix}\Delta P \\ \Delta Q\end{bmatrix} = \begin{bmatrix} H & 0 \\ 0 & L\end{bmatrix}\begin{bmatrix}\Delta\delta \\ \frac{\Delta V}{V}\end{bmatrix}$$

$$\Delta P = H\Delta\delta$$

$$\Delta Q = L\frac{\Delta V}{V}$$

Off-diagonal element of H is –

$$\begin{aligned}
H_{ik} &= \frac{\partial P_i}{\partial Q_k} = V_i V_k Y_{ik} \sin(\theta_{ik} + \delta_i - \delta_k) \\
&= V_i V_k Y_{ik} [\sin\theta_{ik} \cos(\delta_i - \delta_k) + \cos\theta_{ik} \sin(\delta_i - \delta_k)] \\
&= V_i V_k [Y_{ik} \sin\theta_{ik} \cos(\delta_i - \delta_k) + Y_{ik} \cos\theta_{ik} \sin(\delta_i - \delta_k)] \\
&= V_i V_k [-B_{ik} \cos(\delta_i - \delta_k) + G_{ik} \sin(\delta_i - \delta_k)]
\end{aligned}$$

Similarly off-diagonal element of L is –

$$\begin{aligned}
L_{ik} &= \frac{\partial Q_i V_k}{\partial V_k} = V_i V_k Y_{ik} \sin(\theta_{ik} + \delta_i - \delta_k) \\
&= V_i V_k [G_{ik} \sin(\delta_i - \delta_k) - B_{ik} \cos(\delta_i - \delta_k)]
\end{aligned}$$

		$$H_{ik} = L_{ik} = V_i V_k [G_{ik} \sin(\delta_i - \delta_k) - B_{ik} \cos(\delta_i - \delta_k)]$$ The diagonal elements of H are given as – $$H_{ii} = \frac{\partial P_i}{\partial \delta_i} = -\sum_{\substack{k=1 \\ k \neq i}}^{n} V_i V_k Y_{ik} \sin(\theta_{ik} + \delta_i - \delta_k)$$ $$= -Q_i + V_i V_i Y_{ii} \sin\theta_{ii} = -Q_i - V_i^2 B_{ii}$$ $$L_{ii} = \frac{\partial Q_i V_i}{\partial V_i} = 2V_i^2 Y_{ii} \sin\theta_{ii} + \sum_{\substack{k=1 \\ k \neq i}}^{n} V_i V_k Y_{ik} \sin(\theta_{ik} + \delta_i - \delta_k)$$ $$= 2V_i^2 Y_{ii} \sin\theta_{ii} + Q_i - V_i^2 Y_{ii} \sin\theta_{ii} = Q_i + V_i^2 Y_{ii} \sin\theta_{ii} = Q_i - V_i^2 B_{ii}$$ $$\Delta P_i = H \Delta\delta = V_i V_k B'_{ik} \Delta\delta_k$$ $$\text{and } \Delta Q_i = L\frac{\Delta V}{V} = V_i V_k B''_{ik} \frac{\Delta V_k}{V_k}$$ $$\frac{\Delta P_i}{V_i} = B' \Delta\delta$$ $$\text{and } \frac{\Delta Q_i}{V_i} = B'' \Delta V$$
	Q14	a) Derive the Newton Raphson power flow analysis algorithm and gives steps for implementation of the algorithm. b) What are the operating constraints in load flow analysis?
	Ans	$$I_i = \sum_{j=1}^{n} Y_{ij} E_j$$ **In polar form, Rewrite the equation (1) as;** $$I_i = \sum_{j=1}^{n} \lvert Y_{ij}\rvert \lvert E_j\rvert \angle(\theta_{ij} + \delta_j) \quad [\because \text{In polar form } Y_{ij} = \lvert Y_{ij}\rvert \angle\theta_{ij};\ E_j = \lvert E_j\rvert \angle\delta_j]$$ **The complex power at bus (*i*) is given as;** $$P_i - jQ_i = E_i^* I_i$$ **Substitute equation (2) in equation (3);** $$P_i - jQ_i = \lvert E_i\rvert \angle -\delta_i \times \left[\sum_{j=1}^{n} \lvert Y_{ij}\rvert \lvert E_j\rvert \angle(\theta_{ij} + \delta_j)\right]$$ **Equating real and imaginary parts we get;** $$P_i^{(cal)} = \sum_{j=1}^{n} \lvert E_i\rvert \lvert E_j\rvert \lvert Y_{ij}\rvert \cos(\theta_{ij} - \delta_i + \delta_j)$$ $$Q_i^{(cal)} = -\sum_{j=1}^{n} \lvert E_i\rvert \lvert E_j\rvert \lvert Y_{ij}\rvert \sin(\theta_{ij} - \delta_i + \delta_j)$$

$$\begin{bmatrix} \Delta P_2^{(k)} \\ \vdots \\ \Delta P_n^{(k)} \\ \Delta Q_2^{(k)} \\ \vdots \\ \Delta Q_n^{(k)} \end{bmatrix} = \begin{bmatrix} \frac{\partial P_2^{(k)}}{\partial \delta_2} & \cdots & \frac{\delta P_2^{(k)}}{\partial \delta_n} & \frac{\partial P_2^{(k)}}{\partial |E_2|} & \cdots & \frac{\partial P_2^{(k)}}{\partial |E_n|} \\ \vdots & & \vdots & \vdots & & \vdots \\ \frac{\partial P_n^{(k)}}{\partial \delta_2} & \cdots & \frac{\partial P_n^{(k)}}{\partial \delta_n} & \frac{\partial P_n^{(k)}}{\partial |E_2|} & \cdots & \frac{\partial P_n^{(k)}}{\partial |E_n|} \\ \frac{\partial Q_2^{(k)}}{\partial \delta_2} & \cdots & \frac{\partial Q_2^{(k)}}{\partial \delta_n} & \frac{\partial Q_2^{(k)}}{\partial |E_2|} & \cdots & \frac{\partial Q_2^{(k)}}{\partial |E_n|} \\ \vdots & & \vdots & \vdots & & \vdots \\ \frac{\partial Q_n^{(k)}}{\partial \delta_2} & \cdots & \frac{\partial Q_n^{(k)}}{\partial \delta_n} & \frac{\partial Q_n^{(k)}}{\partial |E_2|} & \cdots & \frac{\partial Q_n^{(k)}}{\partial |E_n|} \end{bmatrix} \begin{bmatrix} \Delta \delta_2^{(k)} \\ \vdots \\ \Delta \delta_n^{(k)} \\ \Delta |E_2^{(k)}| \\ \vdots \\ \Delta |E_n^{(k)}| \end{bmatrix}$$

The above equation can be written as;

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J_1 \; J_2 \\ J_3 \; J_4 \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta |E| \end{bmatrix}$$

For J_1:

The diagonal elements are:

$$\frac{\partial P_i}{\partial \delta_j} = \sum_{\substack{j=1 \\ \neq i}}^{n} |E_i||E_j||Y_{ij}| \sin(\theta_{ij} - \delta_i + \delta_j)$$

The off-diagonal elements are:

$$\frac{\partial P_i}{\partial \delta_i} = -|E_i||E_j||Y_{ij}| \sin(\theta_{ij} - \delta_i + \delta_j) \qquad [\because j \neq i]$$

For J_2:

The diagonal elements are:

$$\frac{\partial P_i}{\partial |E_i|} = 2|E_i||Y_{ij}| \cos\theta_{ii} + \sum_{\substack{j=1 \\ \neq i}}^{n} |E_j||Y_{ij}| \cos(\theta_{ij} - \delta_i + \delta_j)$$

The off-diagonal elements are:

$$\frac{\partial P_i}{\partial |E_j|} = |E_i||Y_{ij}| \cos(\theta_{ij} - \delta_i + \delta_j) \qquad [\because j \neq i]$$

For J_3:

The diagonal elements are:

$$\frac{\partial Q_i}{\partial \delta_i} = \sum_{\substack{j=1 \\ \neq i}}^{n} |E_i||E_j||Y_{ij}| \cos(\theta_{ij} - \delta_i + \delta_j)$$

The off-diagonal elements are:

$$\frac{\partial Q_i}{\partial \delta_j} = -|E_i||E_j||Y_{ij}|\cos(\theta_{ij} - \delta_i + \delta_j) \qquad [\because j \neq i]$$

For J_4:

The diagonal elements are:

$$\frac{\partial Q_i}{\partial |E_i|} = -2|E_i||Y_{ii}|\sin\theta_{ii} + \sum_{\substack{j=1\\ \neq i}}^{n} |E_j||Y_{ij}|\sin(\theta_{ij} - \delta_i + \delta_j)$$

The off-diagonal elements are:

$$\frac{\partial Q_i}{\partial |E_j|} = -|E_i|\,|Y_{ij}|\sin(\theta_{ij} - \delta_i + \delta_j) \qquad [\because j \neq i]$$

$$\Delta P_i^{(k)} = P_i^{(sch)} - P_i^{(cal)}$$

$$\Delta Q_i^{(k)} = Q_i^{(sch)} - Q_i^{(cal)}$$

$$\delta_i^{(k+1)} = \delta_i^{(k)} + \Delta\delta_i^{(k)}$$

$$|E_i^{(k+1)}| = |E_i^{(k)}| + \Delta|E_i^{(k)}|$$

b) The operating constraints are including; the injected series power of another line while the last parameter in the second line voltages, the injected line currents passing through the converters would be released to achieve the exchanged power condition and exchanged powers among series converters.

- The load flow studies involves the solution of the power system network under steady state conditions
- The solution is obtained by considering certain inequality constraints imposed on node voltages & reactive power of generators
- The main informations obtained from a load flow study are the magnitude & phase angle of the voltage at each bus & the real & reactive power flowing in each line
- The load flow solution also gives the initial conditions of the system when the transient behaviour of the system is to be studied
- The load flow study of a power system is essential to decide the best operation of the existing system & for planning the future expansion of the system
- It is also essential for designing a new power system

A load flow study of a power system generally require the following steps

- Representation of the system by single line diagram
- Determining the impedance diagram using the informations in single line diagram
- Formulation of network equations
- Solution of network equations

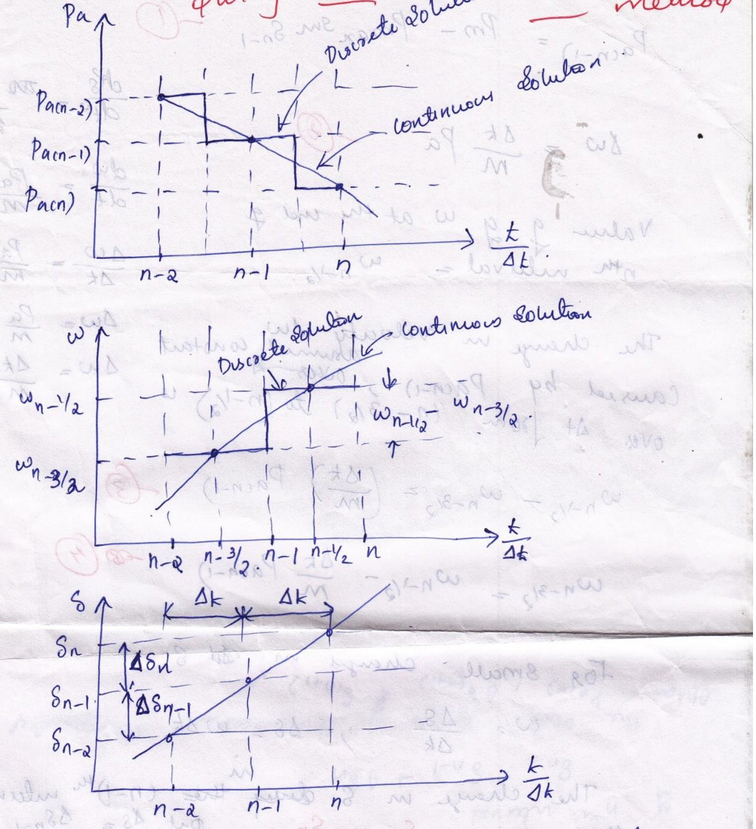
Module 3

	Q15	a) Explain the method of solving swing equation by point-by-point method. b) A generator having H=6.0 MJ/MVA is delivering power of 1p.u. to an infinite bus through a purely reactive network when the occurrence of a fault reduces the generator output power to zero. The maximum power that could be delivered is 2.5per unit. When the fault is cleared the original network conditions again exist. Determine the critical clearance angle and time.

Ans

a)

Swing Equation by point by point method



Point by point solution of swing equation

n represents the discretized time instant.

The numbering on $t/\Delta t$ axis pertains to the end of intervals.

$\delta_0 \Rightarrow$ Angle corresponds to initial operating point

At the end of $(n-1)^{th}$ interval

δ_{n-1}, $P_{a(n-1)}$, $\omega_{n-3/2}$.

Swing equation by point by point pattern

$P_{a(n-1)} = P_m - P_{max} \sin \delta_{n-1}$ — (1)

$\frac{d^2\delta}{dt^2} = \frac{P_a}{M}$

$\Delta \omega = \frac{\Delta t}{M} P_a$ — (2)

$\frac{d\omega}{dt} = \frac{P_a}{M}$

Value of ω at the end of n^{th} interval $= \omega_{n-1/2}$.

$\frac{\Delta\omega}{\Delta t} = \frac{P_a}{M}$

$\Delta\omega = \frac{P_a}{M} \cdot \Delta t$

$\Delta\omega = \frac{\Delta t}{M} P_a$

The change in velocity $\Delta\omega$ caused by $P_{a(n-1)}$, assumed constant over Δt from $(n-3/2)$ to $(n-1/2)$ is

$\omega_{n-1/2} - \omega_{n-3/2} = \left(\frac{\Delta t}{M}\right) P_{a(n-1)}$ — (3)

$\omega_{n-3/2} = \omega_{n-1/2} - \frac{\Delta t}{M} P_{a(n-1)}$ — (4)

For small changes in δ

$\omega = \frac{\Delta\delta}{\Delta t}$; $\Delta\delta = \omega \Delta t$

The change in δ ~~during~~ in the $(n-1)^{th}$ interval

put $\Delta\delta = \Delta\delta_{n-1}$, $\omega = \omega_{n-3/2}$

$\Delta\delta_{n-1} = \delta_{n-1} - \delta_{n-2}$,

$\Delta\delta_{n-1} = \Delta t\, \omega_{n-3/2}$ — (5)

Similarly for change in δ during n^{th} interval

put $\Delta\delta = \Delta\delta_n$ & $\omega = \omega_{n-1/2}$.

$\Delta\delta_n = \delta_n - \delta_{n-1}$

$\Delta\delta_n = \Delta t\, \omega_{n-1/2}$ — (6)

(6) − (5)

$\Delta\delta_n - \Delta\delta_{n-1} = \frac{\Delta t^2}{M} P_{a(n-1)}$

$\Delta\delta_n = \Delta\delta_{n-1} + \frac{\Delta t^2}{M} P_{a(n-1)}$

Let δ_n = the value of δ at the end of n^{th} interval.

$\delta_n = \delta_{n-1} + \Delta\delta_n$

The above process of computation is repeated to obtain $P_{a(n)}$, $\Delta\delta_{(n+1)}$ & $\delta_{(n+1)}$.

b)

$$\delta_{cc} = \cos^{-1}\left[\frac{P_m}{P_{max}}(\delta_{max} - \delta_0) + \cos\delta_{max}\right]$$

$$\frac{H}{\pi f}\frac{d^2\delta}{dt^2} = P_m - P_e$$

During a three phase fault, $P_e = 0$

$$\therefore \frac{H}{\pi f}\frac{d^2\delta}{dt^2} = P_m$$

$$\frac{d^2\delta}{dt^2} = \frac{\pi f P_m}{H}$$

$$\delta = \frac{\pi f}{2H} P_m t^2 + \delta_0$$

where δ_0 is the integral constant

$$\therefore \text{at } t = t_{cc}, \quad \delta_{cc} = \frac{\pi f}{2H} P_m t_{cc}^2 + \delta_0$$

$$\frac{\pi f}{2H} P_m t_{cc}^2 = \delta_{cc} - \delta_0$$

$$\therefore t_{cc} = \sqrt{\frac{2H(\delta_{cc} - \delta_0)}{\pi f P_m}}$$

δ_{cr}= 89.374 = 1.5599 rad

t_{cr}= 0.2704 sec

Q16

a) Write all methods to improve steady state stability limit of power system.

b) A three phase fault is applied at the point P as shown in Fig.2. Find the critical clearing angle for clearing the fault with simultaneous opening of the breakers 1 and 2. The reactance values of various components are indicated in the diagram. The generator is delivering 1.0 p.u. power at the instant preceding the fault.

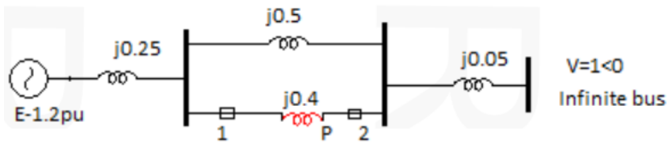


Fig.2

Ans

a) The maximum power transferred between an alternator, and a motor is directly proportional to the product of internal emf of the machines and inversely proportional to the line reactance. The steady state limit is increased because of the two reasons;

- By increasing the excitation of a generator or motor or both – The excitation increase the internal emf and consequently the maximum powered transferred between the two machines increases. Further with the increased value of internal EMFs, the load angle δ decreases.
- Reducing the transfer Reactance – The reactance is reduced by increasing the parallel line between the transmission points. The use of bundle conductor is the other method of reducing the reactance of the line. The reactance can also be decreased by using the capacitance in series with the line.

b)

I. Normal operation (prefault)

$$X_{\mathrm{I}} = 0.25 + \frac{0.5 \times 0.4}{0.5 + 0.4} + 0.05$$

$$= 0.522 \text{ pu}$$

$$P_{e\mathrm{I}} = \frac{|E'|\,|V|}{X_I} \sin\delta = \frac{1.2 \times 1}{0.522} \sin\delta$$

$$= 2.3 \sin\delta$$

Prefault operating power angle is given by

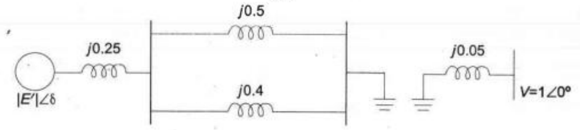
$$1.0 = 2.3 \sin \delta_0$$

or $$\delta_0 = 25.8^\circ = 0.45 \text{ radians}$$

II. During fault

It is clear from Fig. 12.31 that no power is transferred during fault, i.e.,

$$P_{e\mathrm{II}} = 0$$



III. Post fault operation (fault cleared by opening the faulted line)

$$X_{\mathrm{III}} = 0.25 + 0.5 + 0.05 = 0.8$$

$$P_{e\mathrm{III}} = \frac{1.2 \times 1.0}{0.8} \sin\delta = 1.5 \sin\delta \qquad \text{(iii)}$$

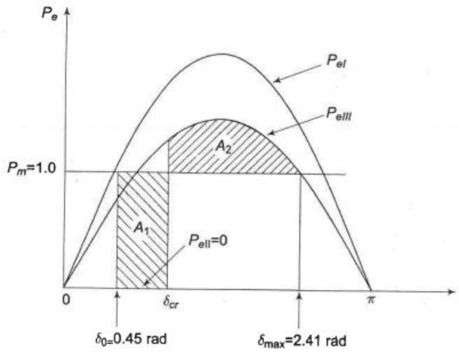


Fig. 12.35

The maximum permissible angle $\delta_{\max}$ for area $A_1 = A_2$ (see Fig. 12.35) is given by

$$\delta_{\max} = \pi - \sin^{-1}\frac{1}{1.5} = 2.41 \text{ radians}$$

Applying equal area criterion for critical clearing angle δ_c

$$A_1 = P_m(\delta_{cr} - \delta_0)$$

$$= 1.0(\delta_{cr} - 0.45) = \delta_{cr} - 0.45$$

$$A_2 = \int_{\delta_{cr}}^{\delta_{max}} (P_{eIII} - P_m)\, d\delta$$

$$= \int_{\delta_{cr}}^{2.41} (1.5 \sin \delta - 1)\, d\delta$$

$$= -1.5 \cos \delta - \delta \Big|_{\delta_{cr}}^{2.41}$$

$$= -1.5 (\cos 2.41 - \cos \delta_{cr}) - (2.41 - \delta_{cr})$$

$$= 1.5 \cos \delta_{cr} + \delta_{cr} - 1.293$$

Setting $A_1 = A_2$ and solving

$$\delta_{cr} - 0.45 = 1.5 \cos \delta_{cr} + \delta_{cr} - 1.293$$

or $\cos \delta_{cr} = 0.843/1.5 = 0.562$

or $\delta_{cr} = 55.8°$

Module 4

Q17 a) Explain the P-f and Q-V control loops of a power system.
b) Draw the block diagram representation of Load Frequency Control (LFC) of a single area system & explain the steady state stability for free governor operation.

Ans a) In a large interconnected system, load frequency equipment is installed for each generator. Similarly for voltage control also voltage control equipment is installed. Fig 2.4 gives the schematic diagram of load frequency and excitation voltage regulators of turbo generators. The controller are set for a particular operating condition and they limit operation in pre-specified limits

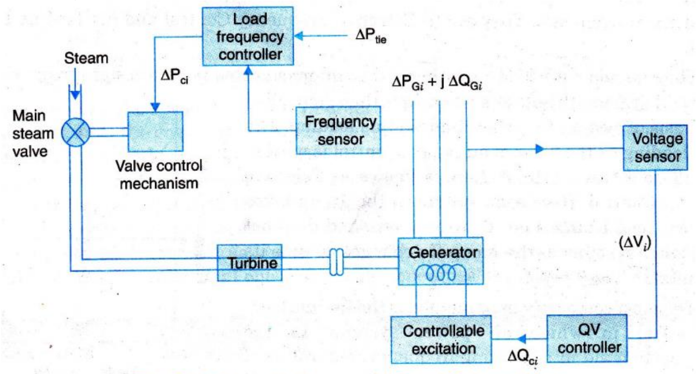


Fig. 20.4 Schematic diagram of L-F and Q-V regulators.

- If the operating conditions change materially the controllers must be re-set either manually or automatically
- It is known that small changes in load depend upon the change in rotor angle δ and is independent of the bus voltage whereas the bus voltage is dependent on machine excitation
- (i.e., on the reactive generation Q, and is independent of rotor angle δ
- Therefore, the two controls, i.e., load frequency and excitation voltage controls are non-interactive for small changes and can be modeled and analyzed independently
- Besides, the load frequency controller is slow acting because of the large time constant contributed by the turbine and generator moment of inertia, and excitation voltage control is fast acting as the time constant of the field winding is relatively smaller, thus the transients in excitation voltage control vanish much faster and do not affect the dynamics of load frequency control

- We will consider here only the load frequency control aspect of regulation
- It is to be noted here that the regulator designed for control should not be insensitive to fast random changes, otherwise the system will be prone to hunting, resulting in excessive wear and tear of control equipments and the rotating machines
- The main objective of the load frequency controller is to exert control of frequency and at the same time of real power exchange via the outgoing lines
- The change in frequency and the tie-line real power are sensed which is a measure of the change in rotor angle δ, i.e., the error $\Delta\delta_i$ to be corrected
- The error signals, i.e., Δf_i and ΔP_{tie} are amplified mixed and transformed into a real power command signal ΔP_{ci} which is sent to the prime mover to call for an increment in the torque
- The prime mover, therefore, brings change in the generator output by an amount ΔP_{Gi} which will change the values of Δf_i and ΔP_{tie}
- The process continues till the deviation Δf_i and ΔP_{tie} are well below the specified tolerances

b) Load Freqency Control

- Let the point A of the speed changer lower down by an amount ΔX_A as a result the commanded increase in power is ΔP_c then $\Delta X_A = K_1\ \Delta P_c$
- The movement of linkage point A causes small position changes ΔX_C & ΔX_D of the linkage points C & D
- With the movement of D upwards by ΔX_D high pressure oil flows into the hydraulic amplifier from the top of the main piston thereby the steam valve will move downwards a small distance ΔX_E which results in increased turbine torque & hence power increase ΔP_G
- This further results in increase in speed & hence frequency of generation
- The increase in frequency Δf causes the link point B to move downward a small distance ΔX_B proportional to Δf
- $\Delta X_C = K_1\Delta f - K_2\,\Delta P_c$ (1)
- The positive constants K_1 & K_2 depend upon the length of the linkage arms AB & BC and upon the proportional constants of the speed changer & the speed governor
- The movement of D is contributed by the movement of C & E
- Since C & E move downwards when D moves upwards,
- $\Delta X_D = K_3\,\Delta X_C + K_4\,\Delta X_E$ (2)
- Assuming that the oil flow into the hydraulic cylinder is proportional to position ΔX_D of the pilot valve, the value of ΔX_E is given by
- $$\mathbf{\Delta X_E = K_5 \int_0^t -(\Delta X_D)dt} \qquad (3)$$

$$\mathbf{\Delta X_C(s) = K_1\Delta F(s) - K_2\Delta P_c(s)} \qquad (4)$$

$$\mathbf{\Delta X_D(s) = K_3\Delta X_C(s) + K_4\Delta X_E(s)} \qquad (5)$$

$$\mathbf{\Delta X_E(s) = -\frac{K_5}{s}\Delta X_D(s)} \qquad (6)$$

- Eliminating the variables ΔX_C & ΔX_D

- $$\mathbf{\Delta X_E(s) = \frac{K_G}{1+sT_G}\left[\Delta P_c(s) - \frac{1}{R}\Delta F(s)\right]} \qquad (7)$$

$R = K_2/K_1$, Speed regulation of the governor

$K_G = K_2 K_3/K_4$ = Gain of speed governor

$T_G = 1/K_4K_5$ = Time constant of speed governor

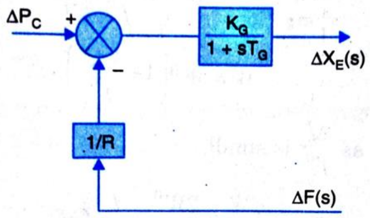


Fig. 20.6 Block diagram representation of speed governing system for steam turbine.

The model requires a relation between changes in power output of the steam turbine to changes in its steam valve opening ΔX_E

Consider a non – reheat turbine with a single gain factor K_T & a single time constant T_T & thus in the crudest model representation of the turbine the transfer function is given as

$$\mathbf{G_T(s) = \frac{\Delta P_t(s)}{\Delta X_E(s)} = \frac{K_T}{1 + sT_T}} \qquad \mathbf{(8)}$$

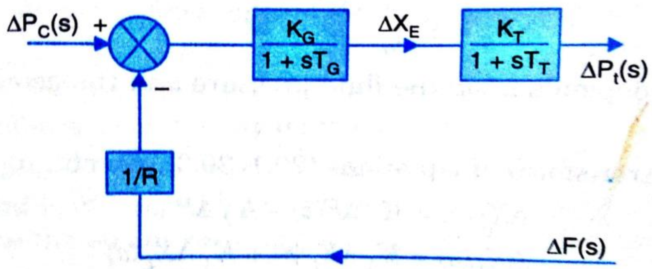


Fig. 20.7 Transfer function representation of power control mechanism of turbine.

Generator – Load Model

- Let W^0 be the KE before the change in load occurs when the frequency is f^0
- Let W be the KE when the frequency is $f^0 + \Delta f$
- Since the KE is proportional to square of the speed of the generator, therefore the two kinetic energies can be correlated as

$$W = W^0\left(\frac{f^0 + \Delta f}{f^0}\right)^2 \qquad (9)$$

$$W = W^0\left(1 + \frac{2\Delta f}{f^0}\right) \qquad (10)$$

neglecting the higher terms as $\Delta f/ f^0$ is small

$$\frac{dW}{dt} = \frac{2W^0}{f^0}\frac{d}{dt}(\Delta f) \qquad (11)$$

$$\left(\frac{\partial P_D}{\partial f}\right)\Delta f = D\Delta f$$

- Therefore, the net power surplus at the bus bar is given by

$$\Delta P_G - \Delta P_D = \frac{2W^0}{f^0}\frac{d}{dt}(\Delta f) + D.\Delta f \qquad (12)$$

- If H is the inertia constant of the generator in MW-sec/MVA & P is the rating in MVA, then $W^0 = HP$

$$\Delta P_G - \Delta P_D = \frac{2HP}{f^0}\frac{d}{dt}(\Delta f) + D.\Delta f \qquad (13)$$

- Dividing throughout by P,

$$\Delta P_{G(p.u)} - \Delta P_{D(p.u)} = \frac{2H}{f^0}\frac{d}{dt}(\Delta f) + D_{(p.u)}\Delta f$$

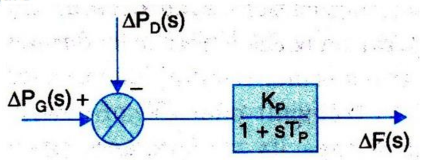
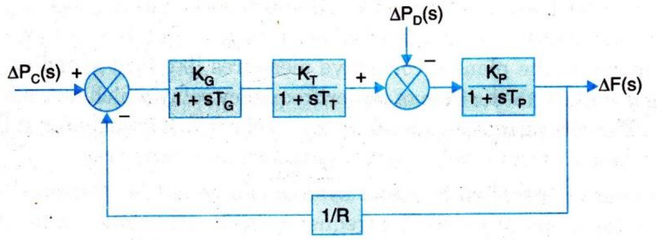
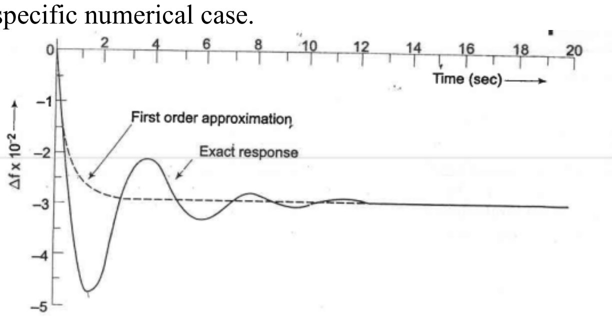
- Taking Laplace transform

$$\Delta P_G(s) - \Delta P_D(s) = \frac{2H}{f^0}s\Delta F(s) + D(s)\Delta F(s)$$

$$\Delta P_G(s) - \Delta P_D(s) = \Delta F(s)\left[\frac{2H}{f^0}s + D(s)\right]$$

$$\Delta F(s) = \frac{\Delta P_G(s) - \Delta P_D(s)}{\frac{2H}{f^0}s + D(s)}$$

$$\Delta F(s) = [\Delta P_G(s) - \Delta P_D(s)]\frac{K_P}{1 + sT_P} \qquad (14)$$

		$T_P = \frac{2H}{Df^0}$ = power system time constant $K_P = \frac{1}{D}$ = power system gain • The block diagram representing eqn (14) is given in figure  **Fig. 20.8** Generator load model.  **Fig. 20.9** Isolated power system load frequency control block diagram.
	Q18	a) Two generators of rating 100 and 200 MW are operated with a droop characteristic of 6% from no load to full load. Determine the load shared by each generator, if a load of 270 MW is connected across the parallel combination of those generators. b) Analyse the dynamic response of frequency for step load change for an isolated power system.
	Ans	a) Percent drop in frequency (or speed) of both machines must be the same: 6x/100 =6(270-x)/200 Load shared by Generator-1 (100 MW unit) = 90 MW and Load shared by Generator-2 (200 MW unit) = 270 – x = 270 – 90 = 180 MW b) To obtain the dynamic response of Load Frequency Control of Isolated Power System giving the change in frequency as function of the time for a step change in load. The characteristic equation being of third order, dynamic response can only be obtained for a specific numerical case.  **Fig. 8.9** Dynamic response of change in frequency for a step change in load (ΔP_D = 0.01 pu, T_{sg} = 0.4 sec, T_t = 0.5 sec, T_{ps} = 20 sec, K_{ps} = 100, R = 3)

$$T_{sg} \ll T_t \ll T_{ps}$$

Typically* $T_{sg} = 0.4$ sec, $T_t = 0.5$ sec and $T_{ps} = 20$ sec.

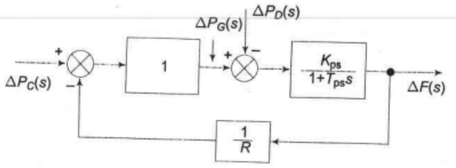


Fig. 8.8 First order approximate block diagram of load frequency control of an isolated area

Letting $T_{sg} = T_t = 0$, (and $K_{sg} K_t \cong 1$), the block diagram of Fig. 8.6 is reduced to that of Fig. 8.8, from which we can write

$$\Delta F(s)|_{\Delta P_C(s)=0} = -\frac{K_{ps}}{(1+K_{ps}/R)+T_{ps}s} \times \frac{\Delta P_D}{s}$$

$$= -\frac{K_{ps}/T_{ps}}{s\left[s+\dfrac{R+K_{ps}}{RT_{ps}}\right]} \times \Delta P_D$$

$$\Delta f(t) = -\frac{RK_{ps}}{R+K_{ps}}\left\{1-\exp\left[-t/T_{ps}\left(\frac{R}{R+K_{ps}}\right)\right]\right\}\Delta P_D \quad (8.22)$$

Taking $R = 3$, $K_{ps} = 1/B = 100$, $T_{ps} = 20$, $\Delta P_D = 0.01$ pu

$$\Delta f(t) = -0.029\,(1 - e^{-1.717t}) \quad (8.23a)$$

$$\Delta f|_{\text{steady state}} = -0.029 \text{ Hz} \quad (8.23b)$$

Module 5

Q19

a) The cost curves of two plants of a generating system are

$\frac{dF_1}{dP_1} = 0.012P_1 + 6.6$ and $\frac{dF_2}{dP_2} = 0.0096P_2 + 6$

The loss coefficient for the above system is given as

$B_{11} = 5\text{x}10\text{-}3$, $B_{12} = B_{21} = -0.03\text{x}10^{-3}$ and $B_{22} = 8\text{x}10^{-3}$ on a base of 100MVA. If plant 1 supplies 200MW and plant 2 supplies 300MW, find the penalty factors of each plant. Is the present dispatch economical? If not, which plant output must be increased and which one decreased? Give reasons also for the same.

b) Distinguish between economic dispatch and unit commitment

Ans

a)

$$\frac{dF_n}{dP_n}.L_n = \lambda$$

where L_n is the penalty factor of plant n and is given by

$$L_n = \frac{1}{1-\dfrac{\partial P_L}{\partial P_n}}$$

$$\simeq \left(1+\frac{\partial P_L}{\partial P_n}\right) \text{ approximate penalty factor}$$

$$= L'_n$$

$$P_n = \frac{1 - \frac{f_n}{\lambda} - \sum_{m \neq n} 2B_{mn}P_m}{\frac{F_{nn}}{\lambda} + 2B_{nn}}$$

$$P_L = \sum_m \sum_n P_n B_{mn} P_m$$

$$\Delta P = |\Sigma P_G - P_L - P_D|$$

L_1=1.02
L_2=1.05
L_1*IC_1=9.12
L_2*IC_2=9.38
As L_1IC_1 is less, its output must be increased.

b)

- Unit commitment and economic dispatch have the same objective of cost minimization. unit commitment has limited access and can minimize the generation/fuel cost it deals with the system operation up to utility end.
  - Unit commitment is a part of economic despatch. It decides the economic operation of partocular unit in a power station. Economic despatch is concerned with economic transmission of energy genetated to load centres economically. Cost of generation is also included in the overall cost in economic despatch.
  - The economic load dispatch is an online process of allocating generation among the available generating units to minimize the total generation cost and satisfy the equality and inequality constraints. Since the civilization increases day by day the demand of electricity increases in the same ratio.
  - Economic dispatch is the short-term determination of the optimal output of a number of electricity generation facilities, to meet the system load, at the lowest possible cost, while serving power to the public in a robust and reliable manner. The problem of unit commitment involves finding the least-cost dispatch of available generation resources to meet the electrical load. Unit commitment aims to make a power system relaiable while economic dispatch uses computatiional approach to determine least cost to meet available load demands.

Q20

a) Define penalty factors and loss coefficients in economic operation of power system.

b) The fuel inputs per hour of plants 1 & 2 are as given:

$F1=0.2P_1^2 +40P_1+120$ Rs/hr
$F2=0.25P_2^2 +30P_2+150$ Rs/hr.

Determine the economic operating schedule and the corresponding cost of generation if the maximum and minimum loading on each unit are 100MW and 25MW and the demand is 180MW and transmission loses are neglected. If the load is equally shared by both the units, determine the savings obtained by loading the units as per the equal incremental production cost.

	Ans	a)

Penalty factor method: From the coordination equation

$$\frac{dF_n}{dP_n} + \lambda \frac{\partial P_L}{\partial P_n} = \lambda$$

or
$$\frac{dF_n}{dP_n} = \lambda\left(1 - \frac{\partial P_L}{\partial P_n}\right)$$

or
$$\frac{dF_n}{dP_n} \cdot \frac{1}{1 - \frac{\partial P_L}{\partial P_n}} = \lambda$$

or
$$\frac{dF_n}{dP_n} . L_n = \lambda$$

where L_n is the penalty factor of plant n and is given by

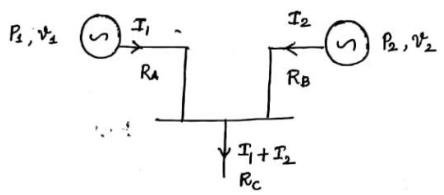
$$L_n = \frac{1}{1 - \frac{\partial P_L}{\partial P_n}}$$

$$\simeq \left(1 + \frac{\partial P_L}{\partial P_n}\right) \text{ approximate penalty factor}$$

$$= L_n'$$

Loss formula using B coefficients.

Consider the single line diagram of a two generating s/m



Let $P_1, V_1, I_1, R_A, P_2, V_2, I_2, R_B, R_C$, be the powers, voltage, current and Resistances of generator 1 and 2 respectively. From the single line diagram, the total transmission loss; P_L is:

$$P_L = 3\left\{ I_1^2 R_A + I_2^2 R_B + (I_1 + I_2)^2 . R_C \right\} \longrightarrow (1)$$

Since this is 3 phase s/m;

$$P_1 = \sqrt{3}\, V_1 I_1 \cos\phi_1 \quad ; \quad P_2 = \sqrt{3}\, V_2 I_2 \cos\phi_2$$

$$\therefore I_1 = \frac{P_1}{\sqrt{3}\, V_1 \cos\phi_1} \quad ; \quad I_2 = \frac{P_2}{\sqrt{3}\, V_2 \cos\phi_2}$$

On substituting the values of I_1 and I_2 in eq^n (1).

$$P_L = 3\left\{ \frac{P_1^2 \cdot R_A}{3\, V_1^2 \cos^2\phi_1} + \frac{P_2^2 \cdot R_B}{3\, V_2^2 \cos^2\phi_2} + \frac{P_1^2 \cdot R_C}{3\, V_1^2 \cos^2\phi_1} + \frac{P_2^2 \cdot R_C}{3\, V_2^2 \cos^2\phi_2} + \frac{2 P_1 P_2 R_C}{3\, V_1 V_2 \cos\phi_1 \cos\phi_2} \right\}$$

$$\Rightarrow P_L = (R_A + R_C) \cdot \frac{P_1^2}{V_1^2 \cos^2\phi_1} + (R_B + R_C) \cdot \frac{P_2^2}{V_2^2 \cos^2\phi_2} + \frac{2\, P_1 . P_2 . R_C}{V_1 . V_2 . \cos\phi_1 . \cos\phi_2}.$$

on re arranging,

$$P_L = P_1 . \underbrace{\left[\frac{R_A + R_C}{V_1^2 \cos^2\phi_1} \right]}_{B_{11}} P_1 + P_2 \underbrace{\left[\frac{R_B + R_C}{V_2^2 \cos^2\phi_2} \right]}_{B_{22}} P_2 + P_1 \underbrace{\left[\frac{R_C}{V_1 V_2 \cos\phi_1 \cos\phi_2} \right]}_{B_{12}} P_2$$

$$+ P_2 \underbrace{\left[\frac{R_C}{V_1 V_2 \cos\phi_1 \cos\phi_2} \right]}_{B_{21}} P_1 .$$

$$\therefore \quad P_L = P_1\, B_{11}\, P_1 + P_2\, B_{22}\, P_2 + P_1\, B_{12}\, P_2 + P_2\, B_{21}\, P_1$$

where B_{11}, B_{22}, B_{12} and B_{21} are β coefficients

$\therefore$ In general; $P_L = \sum_m \sum_n P_m B_{mn} P_n$; which is the loss formula.

$\therefore$ For a two generator s/m.

$P_L = \sum_m \sum_n P_m B_{mn} P_n$: where m and n varies from 1 to 2.

$$\therefore \quad P_L = P_1\, B_{11}\, P_1 + P_1\, B_{12} . P_2 + P_2\, B_{22}\, P_2 + P_2\, B_{21}\, P_1$$

b)

Solution: The incremental production costs of both the units are

$$\frac{dF_1}{dP_1} = 0.4\, P_1 + 40 \text{ Rs. per MWhr}$$

and $$\frac{dF_2}{dP_2} = 0.5\, P_2 + 30 \text{ Rs. per MWhr}$$

Now for economic operation of the units

$$\frac{dF_1}{dP_1} = \frac{dF_2}{dP_2}$$

i.e., $$0.4P_1 + 40 = 0.5P_2 + 30$$

and $$P_1 + P_2 = 180$$

Solution of these equations gives

$$P_1 = 88.89 \text{ MW} \quad \text{and} \quad P_2 = 91.11 \text{ MW}$$

Now cost of generation = $F_1 + F_2$

$$F_1 = 0.2\, P_1^2 + 40\, P_1 + 120 = \text{Rs. } 5255.88/\text{hr.}$$

$$F_2 = 0.25\, P_2^2 + 30\, P_2 + 150 = \text{Rs. } 4958.55/\text{hr.}$$

Total cost = Rs. 10214.43/hr.

(*b*) If the load on each unit is 90 MW, the cost of generation will be

$$F_1 = \text{Rs. } 5340/\text{hr.}$$

$$F_2 = \text{Rs. } 4875/\text{hr.}$$

Total cost = Rs. 10215/hr.

∴ Saving will be Rs. 0.57/hr.

Savings= Rs0.57/hr
=0.57*24 = 13.68 Rs/day
=4993 Rs/year